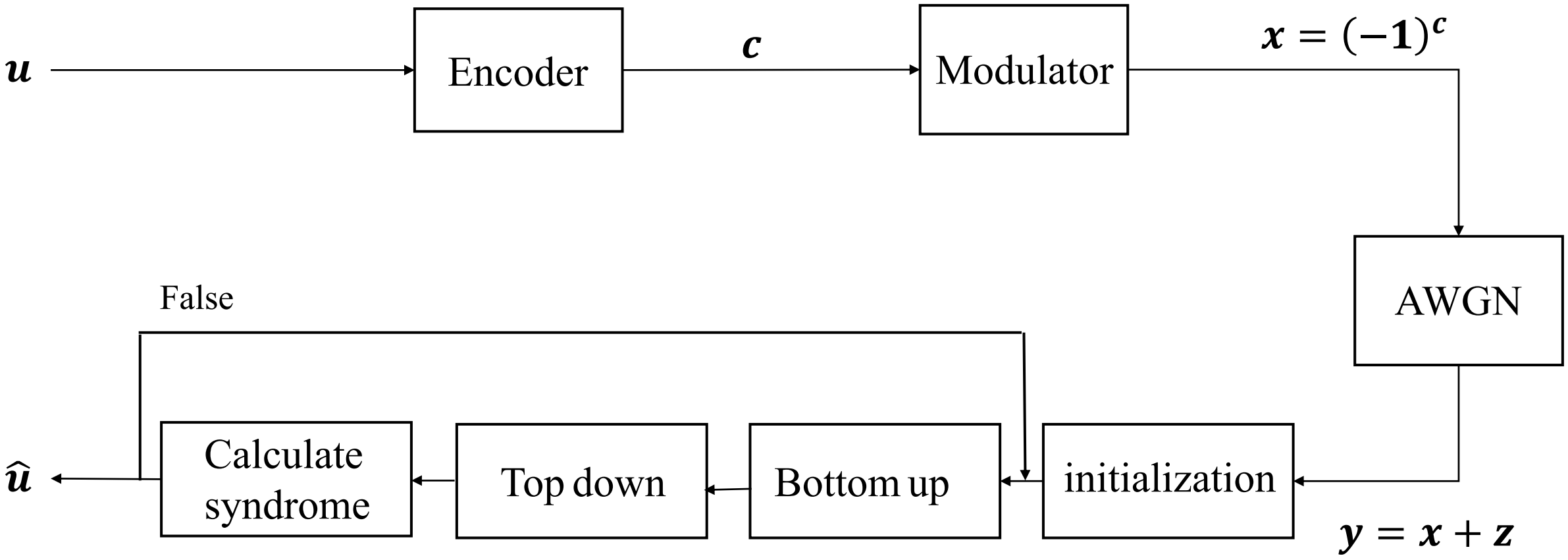
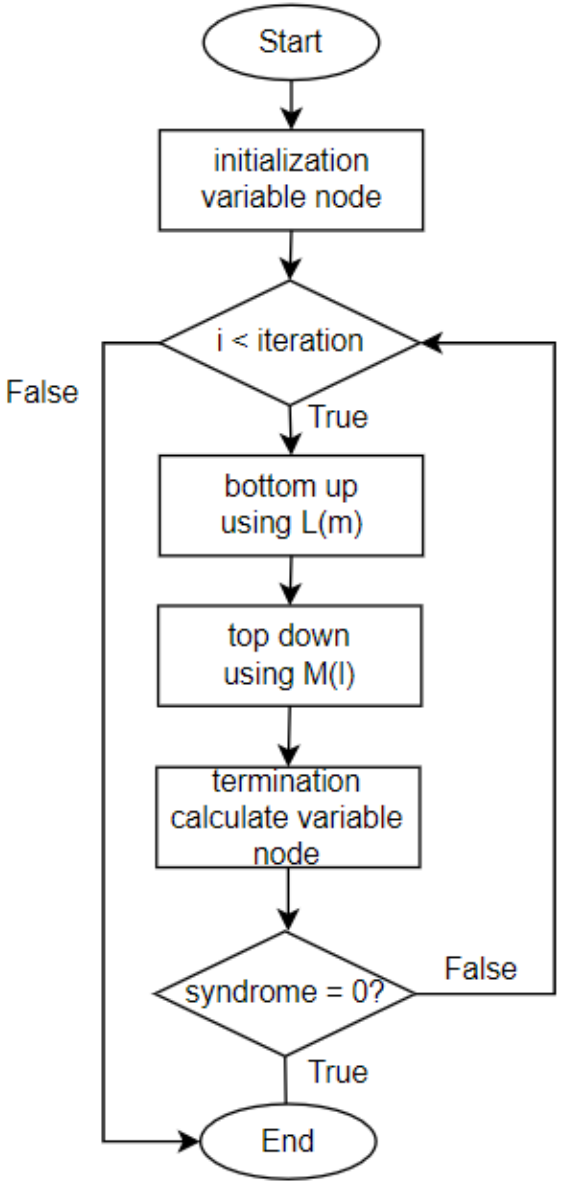
LDPC Decoding

E24104024 李姿穎

Block Diagram



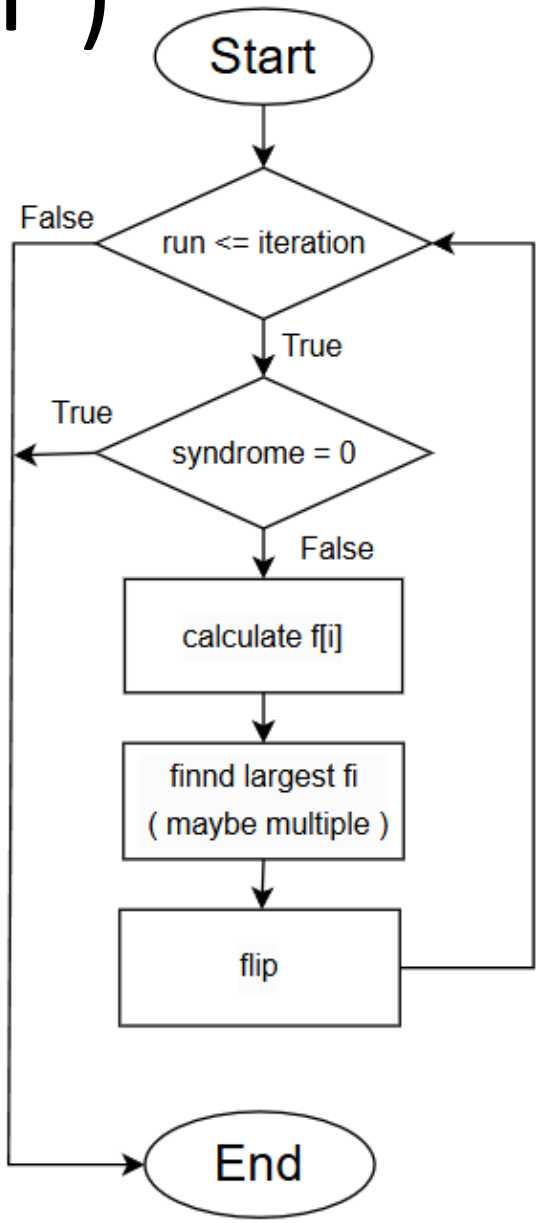
Flow Chart (for SPA and MSA)



- Min-Sum Algorithm(MSA)
  $CHK(L_1, L_2) = \mathrm{sgn}(L_1)\,\mathrm{sgn}(L_2)\min(|L_1|, |L_2|)$

- Sum-Product Algorithm(SPA)
  $CHK(L_1, L_2) = 2 * \tanh\text{^}{-1}\,(\tanh(\frac{L1}{2}) * \tanh(\frac{L2}{2}))$

Flow Chart (for BF)



Data (for SPA)

Parameter	Value
iteration	100
Seed	-2024

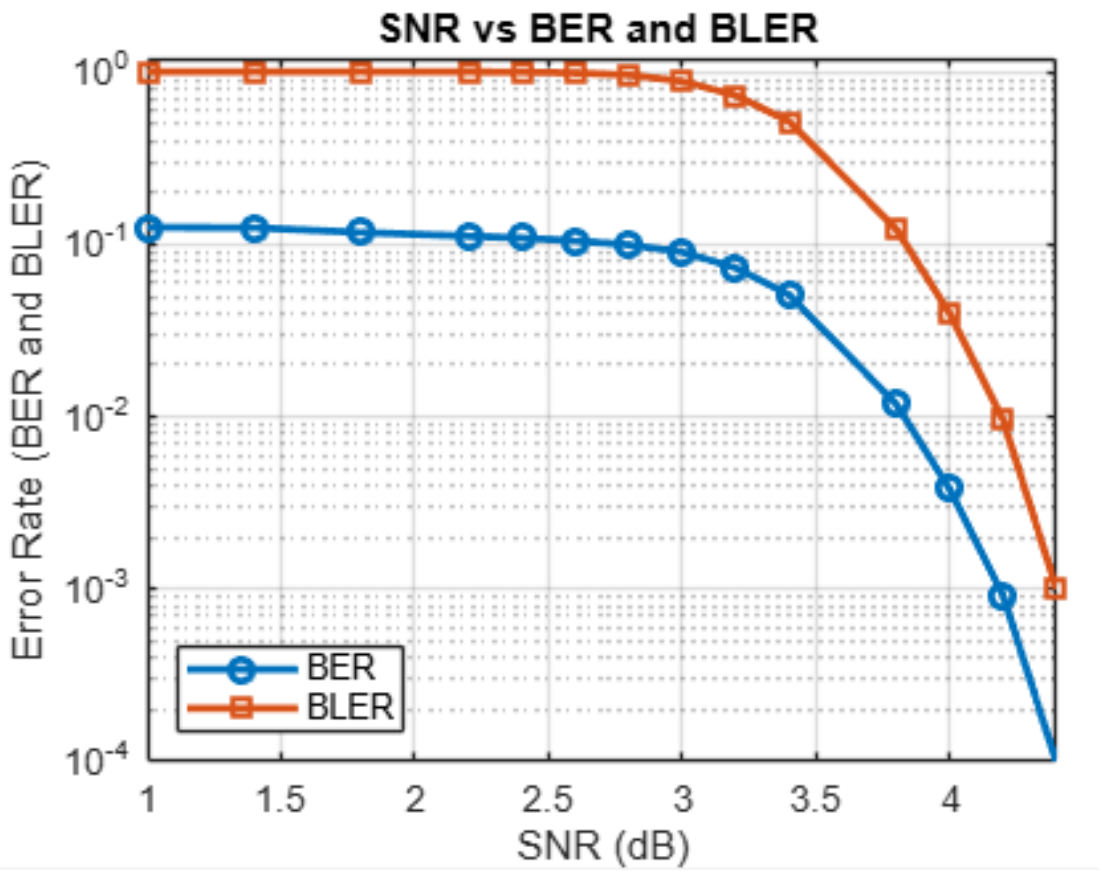
參考數值：

- Sum-Product Algorithm (Maximum 100 iterations):
  - BER = 1.75×10^{-2} and BLER = 4.27×10^{-1} at $E_b/N_0 = 2.4$ dB;
  - BER = 1.04×10^{-3} and BLER = 2.41×10^{-2} at $E_b/N_0 = 3.0$ dB;
  - BER = 7.48×10^{-5} and BLER = 1.88×10^{-3} at $E_b/N_0 = 3.4$ dB;

SNR	1.0	1.4	1.8	2.2	2.4	2.6	2.8	3.0
Decoded Block	100	100	100	1000	1000	1000	1000	1000
BER	0.082177	0.072036	0.058387	0.03842	0.023735	0.011544	0.004988	0.001352
BLER	1	1	0.93	0.684	0.437	0.216	0.095	0.025
Average iteration	100	100	93.68	71.438	48.757	27.555	15.166	7.513

SNR	3.2	3.4	3.8	4.0	4.2	4.4
Decoded Block	1000	10000	10000	10000	10000	10000
BER	0.000411	0.000062	0.000004	0	0	0
BLER	0.007	0.0011	0.0001	0	0	0
Average iteration	4.091	2.701	1.9844	1.784	1.5815	1.3596

Simulation Diagram (for SPA)



Data (for MSA)

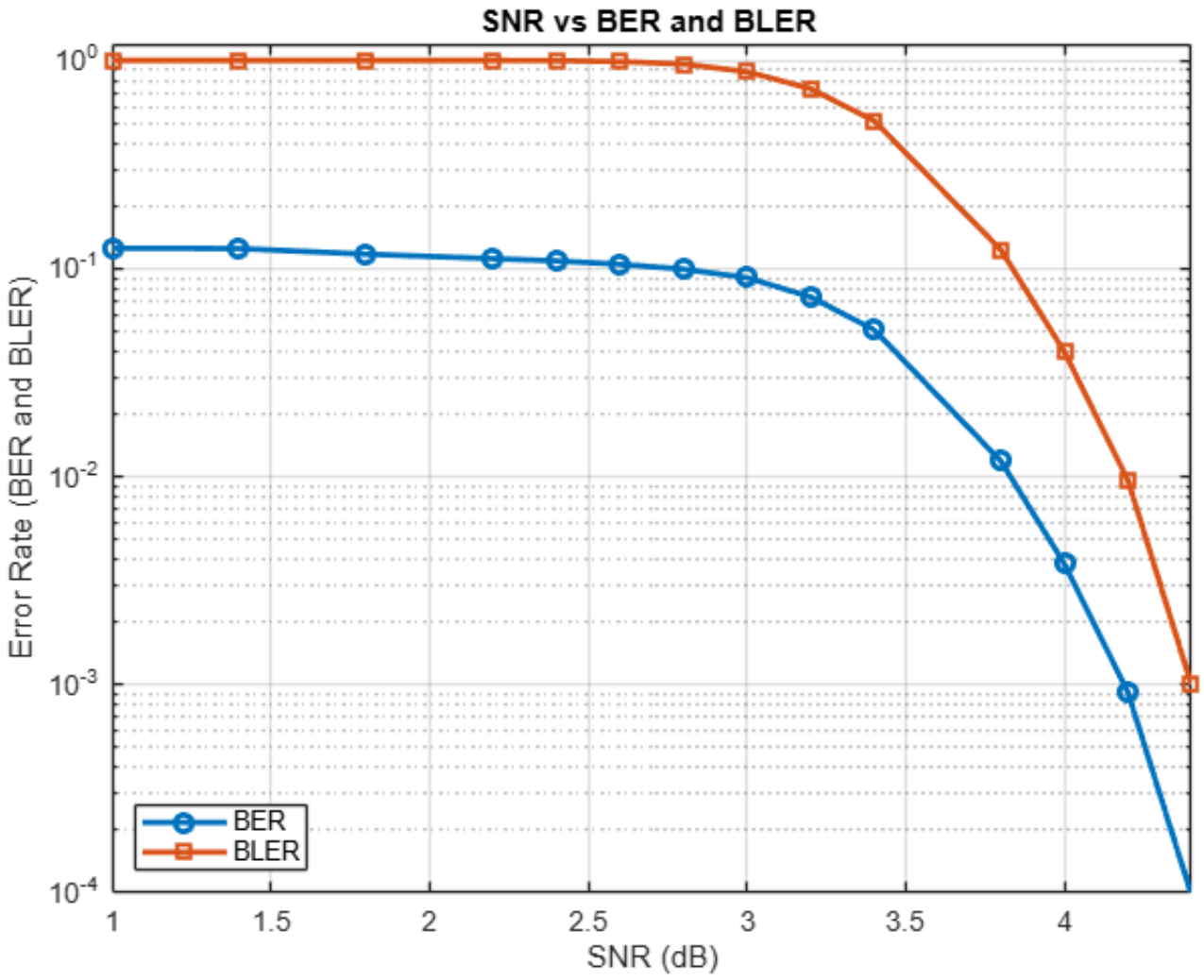
Parameter	Value
iteration	100
Seed	-2024

- Min-Sum Algorithm (Maximum 100 iterations):
  - BER = 9.14×10^{-4} and BLER = 9.73×10^{-3} at $E_b/N_0 = 4.2$ dB;
  - BER = 1.34×10^{-4} and BLER = 1.52×10^{-3} at $E_b/N_0 = 4.4$ dB;

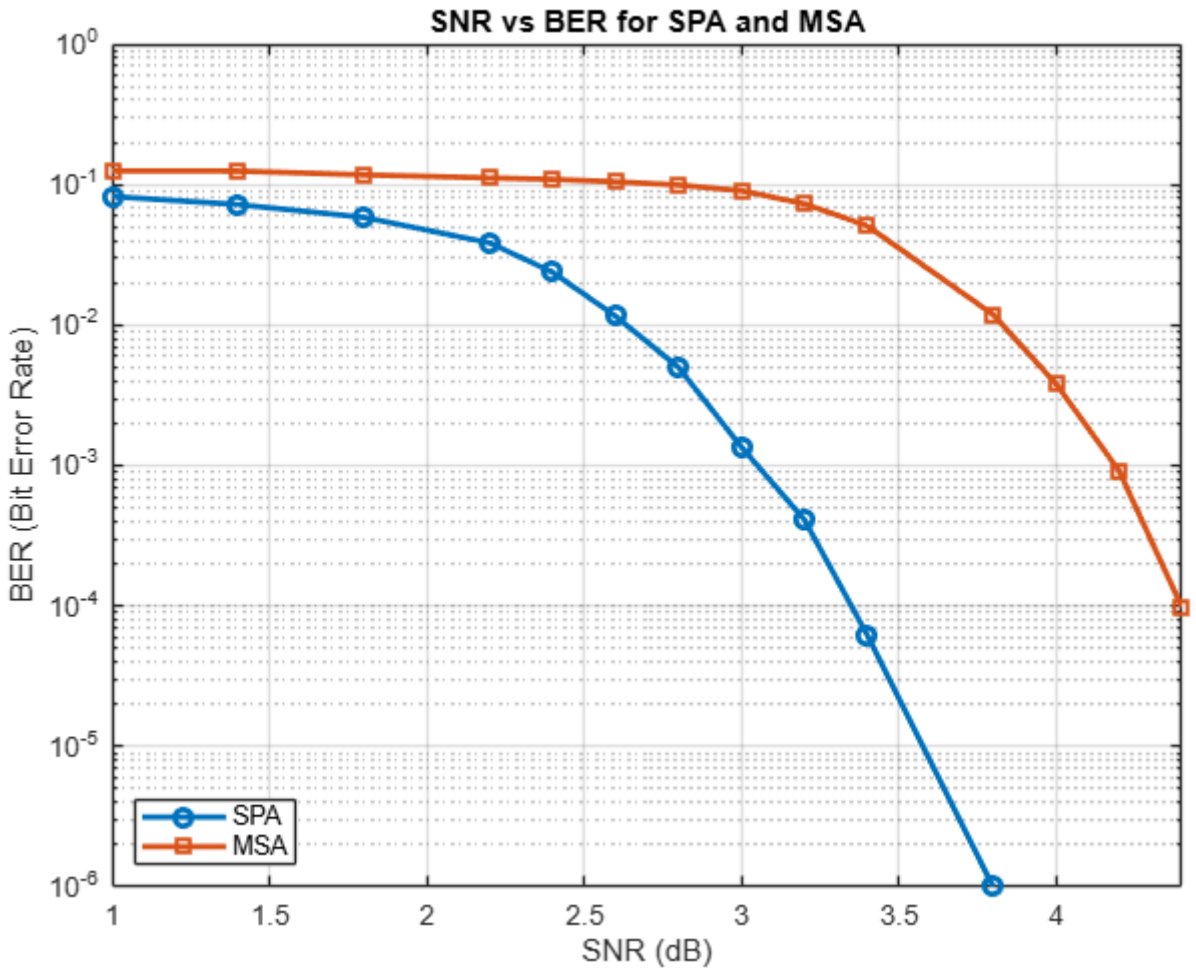
SNR	1.0	1.4	1.8	2.2	2.4	2.6	2.8	3.0
Decoded Block	100	100	100	1000	1000	1000	1000	1000
BER	0.125198	0.124699	0.117183	0.111471	0.10883	0.104702	0.099237	0.090312
BLER	1	1	1	1	0.997	0.987	0.96	0.882
Average iteration	100	100	100	100	99.709	98.739	96.109	88.523

SNR	3.2	3.4	3.8	4.0	4.2	4.4
Decoded Block	1000	10000	10000	10000	10000	10000
BER	0.073014	0.050811	0.011861	0.003807	0.000919	0.000098
BLER	0.731	0.514	0.1228	0.0398	0.0096	0.001
Average iteration	73.797	52.586	14.0989	5.7704	2.5871	1.5069

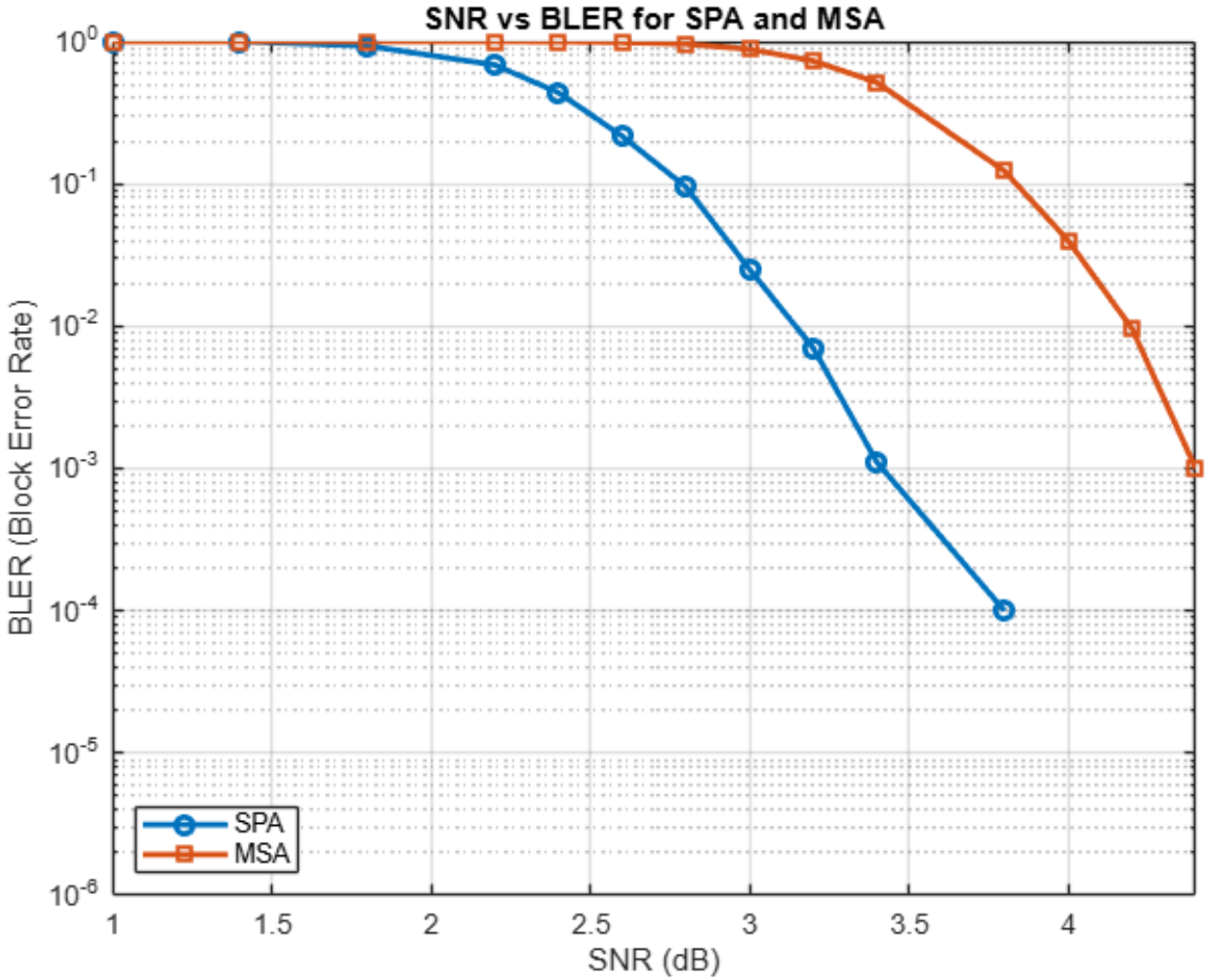
Simulation Diagram (for MSA)



Simulation Diagram
(BER : SPA compare MSA)



Simulation Diagram
(BER : SPA compare MSA)



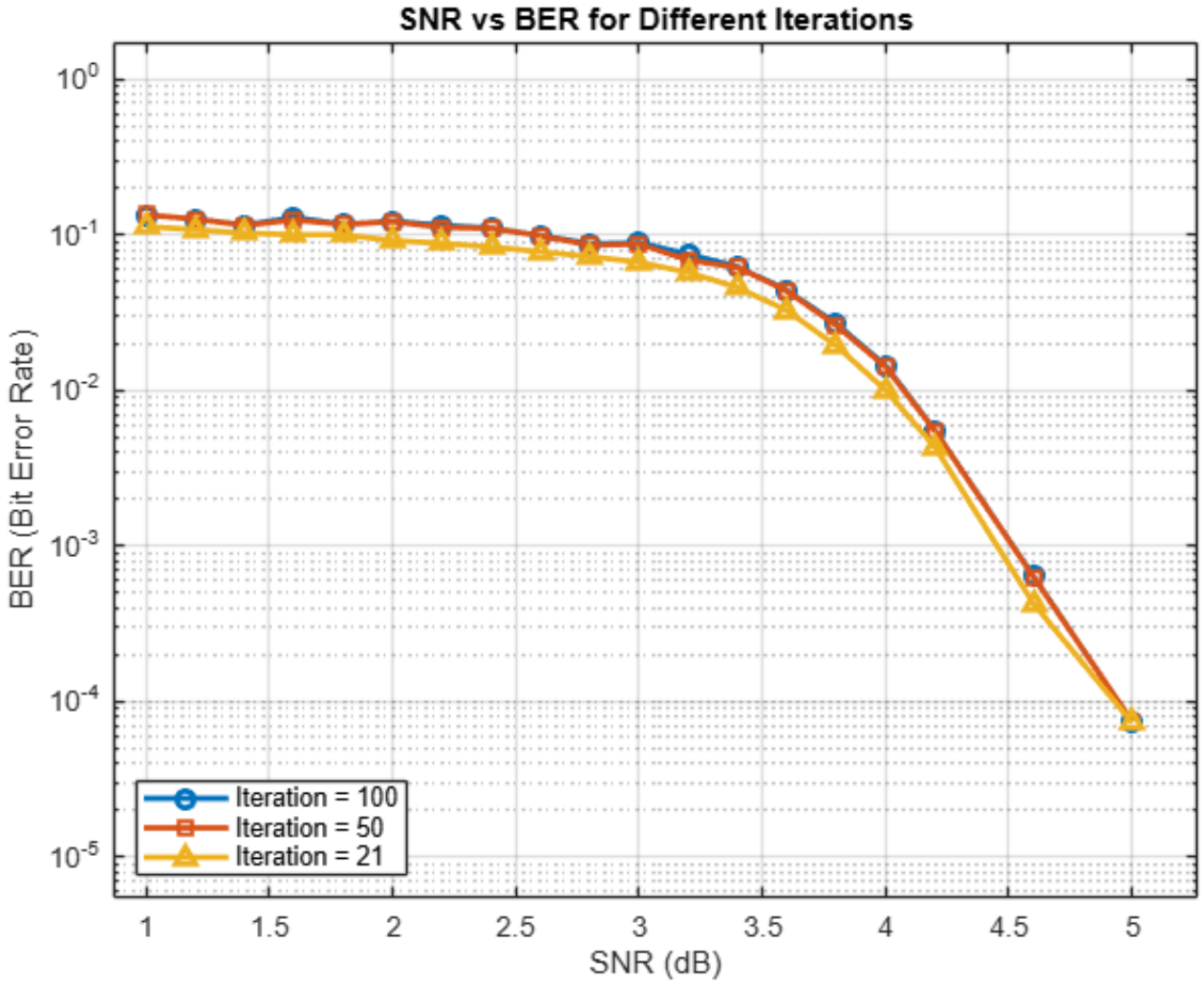
Data (BF with different iteration)

Parameter	Value
Seed	-2024

SNR	1.0	1.2	1.4	1.6	1.8	2.0	2.2	2.4	2.6	2.8	3.0
Iteration =100 BER	0.133534	0.125736	0.114942	0.128649	0.11605	0.121951	0.114115	0.109945	0.098298	0.087037	0.088235
Iteration =50 BER	0.133636	0.125659	0.114379	0.124245	0.11589	0.121027	0.111012	0.109606	0.09798	0.085442	0.086732
Iteration =21 BER	0.112561	0.107357	0.102087	0.099616	0.099661	0.091202	0.087789	0.083138	0.077483	0.072145	0.066322

SNR	3.2	3.4	3.6	3.8	4.0	4.2	4.6	5.0
Iteration =100 BER	0.074566	0.06197	0.04365	0.026641	0.014349	0.0055	0.000634	0.000074
Iteration =50 BER	0.068327	0.061414	0.043195	0.026017	0.014112	0.005473	0.000629	0.000074
Iteration =21 BER	0.05701	0.045649	0.032396	0.019244	0.009992	0.004304	0.000424	0.000073

Simulation Diagram
(BF with different iteration)



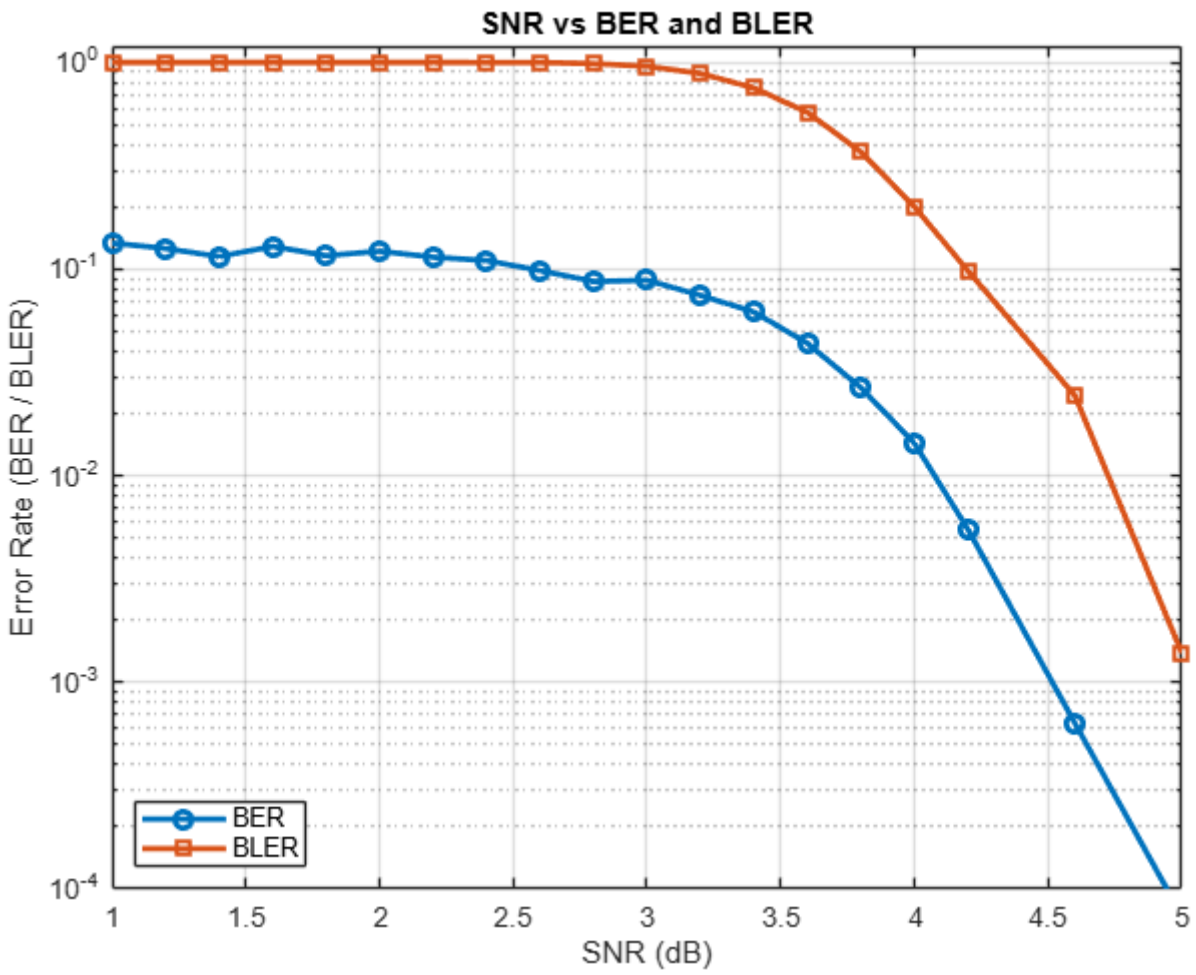
Data (for BF)

Parameter	Value
iteration	100
Seed	-2024

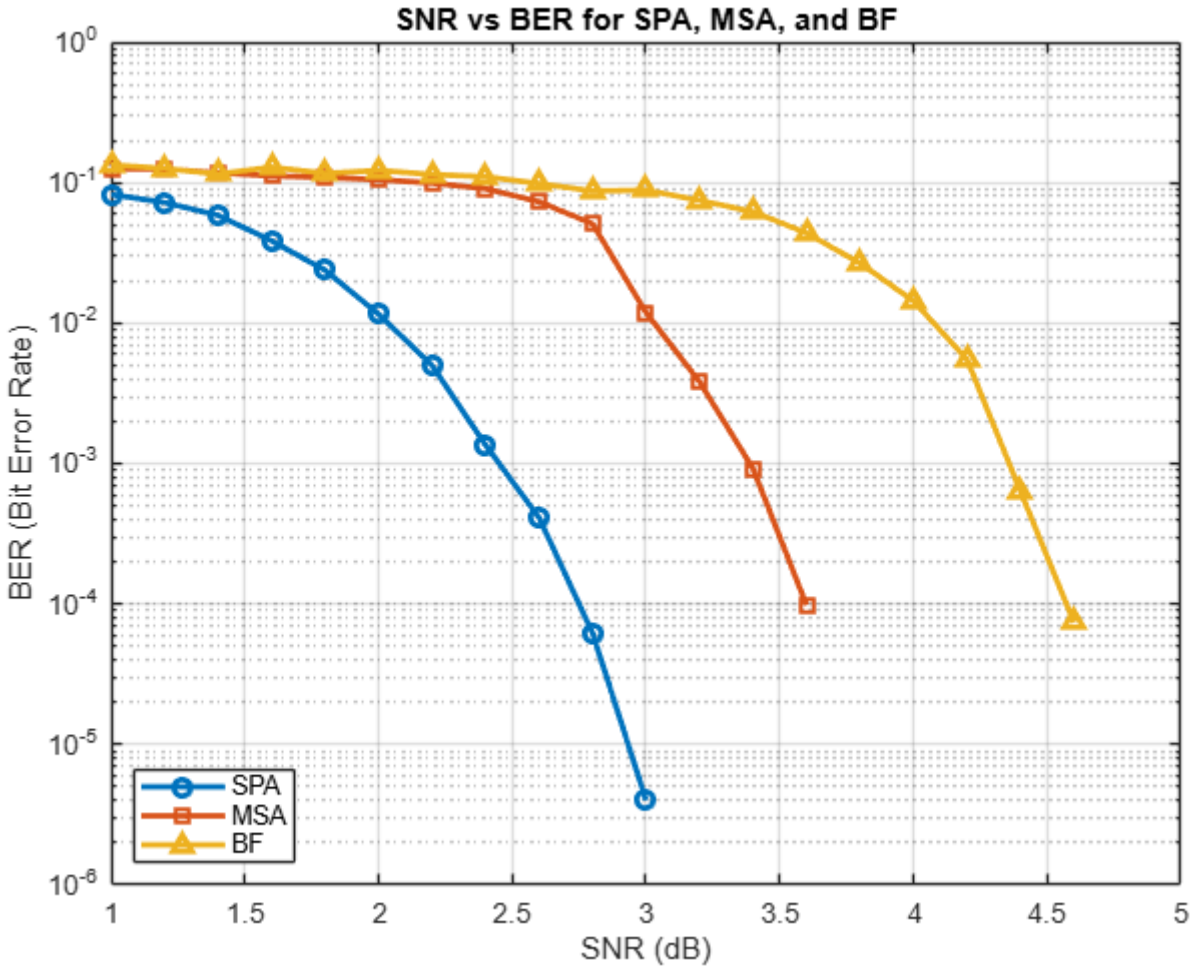
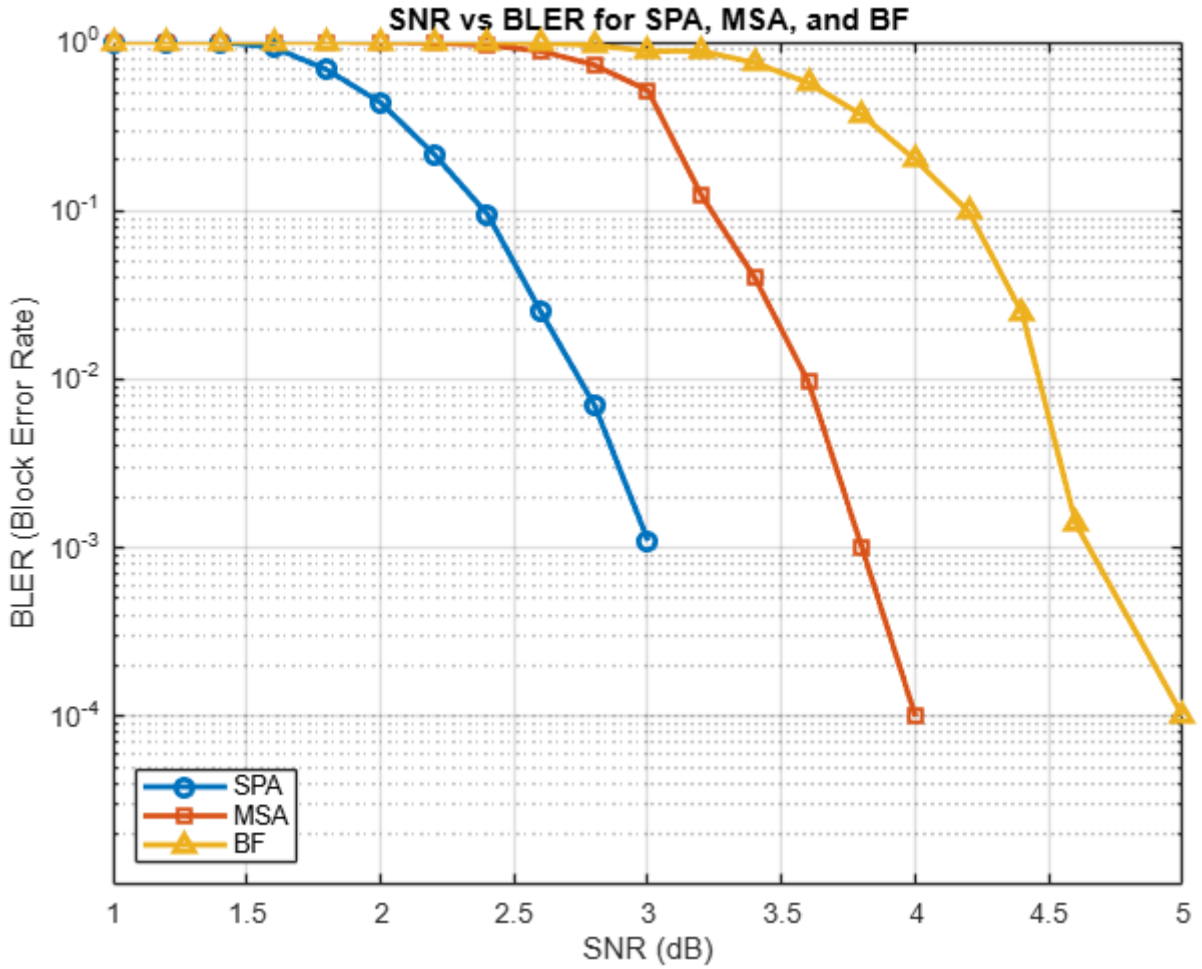
SNR	1.0	1.2	1.4	1.6	1.8	2.0	2.2	2.4	2.6	2.8	3.0
Decoded Block	100	100	100	200	200	1000	1000	1000	1000	1000	10000
BER	0.133534	0.125736	0.114942	0.128649	0.11605	0.121951	0.114115	0.109945	0.098298	0.087037	0.088235
BLER	1	1	1	1	1	1	1	0.999	0.999	0.987	0.9571
Average iteration	93.55	97.97	96.8	93.55	94.975	94.94	96.427	95.066	95.066	95.426	91.996

SNR	3.2	3.4	3.6	3.8	4.0	4.2	4.6	5.0
Decoded Block	10000	10000	10000	10000	10000	10000	10000	50000
BER	0.074566	0.06197	0.04365	0.026641	0.014349	0.0055	0.000634	0.000074
BLER	0.8844	0.7519	0.5725	0.3686	0.2	0.0981	0.0244	0.001384
Average iteration	86.2428	75.1617	59.6204	42.065	27.0404	17.5334	9.4136	6.86724

Simulation Diagram (for BF)



Simulation Diagram
(BER : SPA & MSA & BF)



Simulation Diagram
(BLER : SPA & MSA & BF)